Leakage-Aware Deep Learning for Financial Event-State Forecasting

Jose Ignacio Valdivia Aguero
University of Luxembourg
Luxembourg

ABSTRACT

This project studies aligned company-news event states for multi-horizon excess-return forecasting beyond strong market baselines. I built a local, reproducible artifact pipeline for the top-50 liquid U.S. equities, with same-day-as-close news alignment, no-news states, purged expanding folds, an untouched lockbox, fold-local preprocessing, saved predictions, and tuning ledgers. The primary XGBoost configuration completed 800 development trials and is the strongest lockbox error model: weighted error improved by -5.34×10^{-4} versus the prior XGBoost baseline with a 62.2% both-metric improvement rate. TSMixer is the strongest lockbox portfolio-ranking model, reaching 15.05% over the 137-day lockbox simulation. The Mixed model improved validation by -2.97×10^{-4} versus XGBoost, but worsened lockbox error by $+9.20\times10^{-5}$. The result is objective-specific: XGBoost wins calibrated error forecasting, while TSMixer wins ranking-driven portfolio simulation.

1 INTRODUCTION

Financial news may contain useful short-lived information, but return prediction is low signal-to-noise and highly vulnerable to leakage, multiple testing, and post-hoc narratives. The project therefore frames the task as end-of-day forecasting of forward excess returns for horizons $\{1, 5, 10, 20, 60\}$ days. The contribution is a reproducible deep-learning-ready benchmark: top-50 equities, Alpha Vantage news, Twelve Data OHLCV, explicit no-news rows, purged walk-forward development folds, a final lockbox, and artifact-backed model comparisons. The implemented roster spans linear and tree market baselines, FinBERT feature sidecars, TSMixer sequence models, XGBoost tuning, and horizon-level XGBoost/TSMixer mixing. AUC is intentionally not reported: the target is continuous excess return, so RMSE/MAE, zero-baseline R^2, rank IC, bootstrap support, and lockbox deltas are the relevant metrics.

2 RELATED WORK

The design follows market-efficiency and asset-pricing principles from Fama, Fama-French, and Campbell-Lo-MacKinlay [4, 7, 8]: small apparent edges require strict out-of-sample controls. Textual-finance work shows why news can matter but also why representation matters [15, 17]. Modern NLP motivates contextual encoders and FinBERT-style finance adaptation [2, 6]. For tabular market features, gradient boosting remains a strong baseline [5, 12]. For sequence modeling, the relevant state of the art includes interpretable multi-horizon attention, patch-based transformers, inverted transformers, and foundation time-series models [1, 13, 14, 16]. Evaluation follows forecasting-metric and time-series validation guidance [3, 11]; multiple-testing control follows Harvey et al. [10]. ROC-AUC is cited only to justify exclusion for this regression target [9].

Table 1: Protocol: Leakage-aware artifact pipeline.

Stage	Artifact	Key control	Output
Data	Top-50 + OHLCV + news	no raw overwrite	row universe
Features	market + event states	no-news rows preserved	sidecars
Splits	expanding folds + lockbox	purge/embargo	split manifest
Training	fold-local preprocessing	no lockbox tuning	saved preds
Evaluation	RMSE/MAE + rank IC	bootstrap + lockbox	reports

Horizons: 1d, 5d, 10d, 20d, 60d. Lockbox ends 2026-02-13; all result roots keep manifests and prediction packets.

3 EXPERIMENTS

Data and splits. The benchmark uses 50 liquid equities, five horizons, forward leave-one-out excess-return labels, and explicit event-state rows for no-news dates. The enhanced Alpha Vantage rebuild aligned 166,711 news rows while preserving provider-truncation limitations. The split protocol defines nine expanding walk-forward development folds plus a held-out lockbox from 2025-09-30 to 2026-02-13; lockbox rows are never used for tuning.

Models and training. XGBoost is the primary event-market candidate, selected from the full 800-trial Optuna search. Its short-horizon portfolio rankings are flat for the 1d and 5d simulations, which identifies a concrete signal-construction issue for the next portfolio simulation. The TSMixer branch remains a frozen sequence-model comparator, and the Mixed model is a lightweight frozen-base horizon mixer over XGBoost and TSMixer predictions; no TSMixer weights were retrained.

Results. Figure 1 shows the key pattern. XGBoost is the strongest active model on lockbox error metrics: per-horizon RMSE is 0.0214, 0.0468, 0.0661, 0.0943, and 0.1929 for 1d, 5d, 10d, 20d, and 60d respectively (Appendix Table 2). TSMixer is the strongest lockbox ranking model in the portfolio simulation: every best-by-horizon portfolio row is TSMixer, including 15.05% at 1d and 12.43% at 60d (Appendix Table 3). The Mixed model improved validation fit but did not beat XGBoost on lockbox error. Thus, error forecasting and cross-sectional ranking produce different winners.

4 DISCUSSION

The strongest lesson is that financial deep learning must be evaluated as a full forecasting system, not as an isolated model leaderboard. XGBoost is the best calibrated error forecaster because it improves the prior XGBoost lockbox weighted error and clears more than half of RMSE/MAE comparisons. TSMixer is the best ranking forecaster because its sequence predictions translate most strongly into lockbox portfolio simulations. The Mixed model adds scientific value by showing that strong validation performance does not automatically translate to lockbox performance. Next steps are to try a deeper reinforcement learning model with capacity to rank

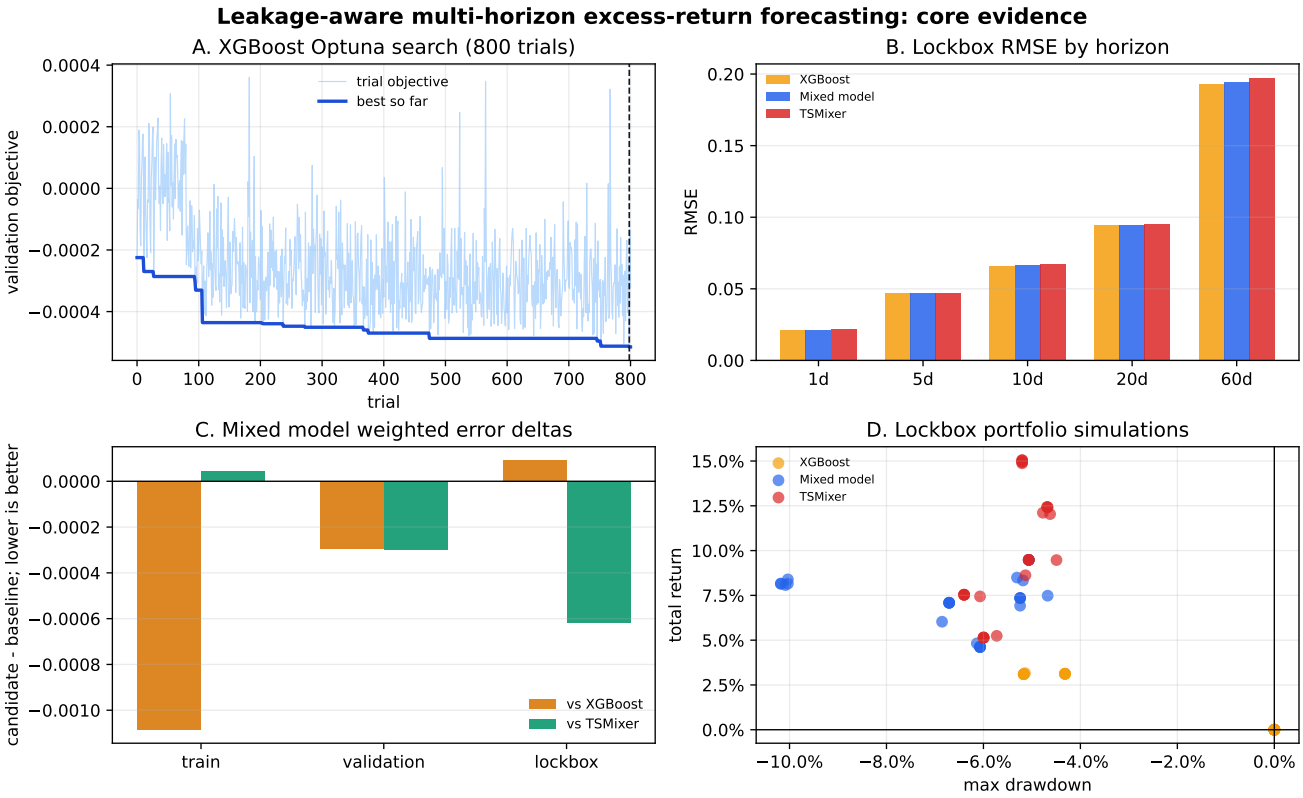


Figure 1: Core evidence. A: XGBoost tuning used train/validation-only selection. B: lockbox RMSE by horizon. C: Mixed model validation gains did not transfer to lockbox. D: the lockbox portfolio simulation summarizes realized forecast rankings over the 137-day evaluation window.

predictions from most probable to least probable, expand data collection beyond the 25-call/day API bottleneck, and add transaction costs for richer portfolio evaluation. The project therefore produces a strong excess-return forecasting benchmark, an objective-specific model conclusion, and a concrete roadmap for improving the next model family.

REFERENCES

[1] Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda-Arango, Shubham Kapoor, Jasper Zschiegner, Danielle C. Maddix, Michael W. Mahoney, Kari Torkkola, Andrew Gordon Wilson, Michael Bohlke-Schneider, and Yuyang Wang. 2024. Chronos: Learning the Language of Time Series. Transactions on Machine Learning Research. https://openreview.net/forum?id=gerNCVqqtR

[2] Dogu Araci. 2019. FinBERT: Financial Sentiment Analysis with Pre-trained Language Models. arXiv:1908.10063 [cs.CL] https://arxiv.org/abs/1908.10063

[3] Christoph Bergmeir, Rob J. Hyndman, and Bonsoo Koo. 2018. A Note on the Validity of Cross-validation for Evaluating Autoregressive Time Series Prediction. *Computational Statistics and Data Analysis* 120 (2018), 70–83. https://doi.org/10.1016/j.csda.2017.11.003

[4] John Y. Campbell, Andrew W. Lo, and A. Craig MacKinlay. 1997. *The Econometrics of Financial Markets*. Princeton University Press, Princeton, NJ, USA.

[5] Tianqi Chen and Carlos Guestrin. 2016. XGBoost: A Scalable Tree Boosting System. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. ACM, New York, NY, USA, 785–794. https://doi.org/10.1145/2939672.2939785

[6] Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. In *Proceedings of NAACL-HLT*. Association for Computational Linguistics, Minneapolis, Minnesota, 4171–4186. https://doi.org/10.18653/v1/N19-1423

[7] Eugene F. Fama. 1970. Efficient Capital Markets: A Review of Theory and Empirical Work. *The Journal of Finance* 25, 2 (1970), 383–417. https://doi.org/10.2307/2325486

[8] Eugene F. Fama and Kenneth R. French. 1993. Common Risk Factors in the Returns on Stocks and Bonds. *Journal of Financial Economics* 33, 1 (1993), 3–56. https://doi.org/10.1016/0304-405X(93)90023-5

[9] Tom Fawcett. 2006. An Introduction to ROC Analysis. *Pattern Recognition Letters* 27, 8 (2006), 861–874. https://doi.org/10.1016/j.patrec.2005.10.010

[10] Campbell R. Harvey, Yan Liu, and Heqing Zhu. 2016. … and the Cross-Section of Expected Returns. *The Review of Financial Studies* 29, 1 (2016), 5–68. https://doi.org/10.1093/rfs/hhv059

[11] Rob J. Hyndman and Anne B. Koehler. 2006. Another Look at Measures of Forecast Accuracy. *International Journal of Forecasting* 22, 4 (2006), 679–688. https://doi.org/10.1016/j.ijforecast.2006.03.001

[12] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. 2017. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. In *Advances in Neural Information Processing Systems*. Curran Associates, Inc., Red Hook, NY, USA, 3146–3154.

[13] Bryan Lim, Sercan O. Arik, Nicolas Loeff, and Tomas Pfister. 2021. Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting. *International Journal of Forecasting* 37, 4 (2021), 1748–1764. https://doi.org/10.1016/j.ijforecast.2021.03.012

[14] Yong Liu, Haixu Wu, Jianmin Wang, and Mingsheng Long. 2024. iTransformer: Inverted Transformers Are Effective for Time Series Forecasting. In *International Conference on Learning Representations*. OpenReview.net, Online, 1 pages. https://openreview.net/forum?id=JePfAI8fah

[15] Tim Loughran and Bill McDonald. 2011. When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks. *The Journal of Finance* 66, 1 (2011), 35–65. https://doi.org/10.1111/j.1540-6261.2010.01625.x

[16] Yuqi Nie, Nam H. Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam. 2023. A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. In *International Conference on Learning Representations*. OpenReview.net, Online, 1 pages. https://openreview.net/forum?id=Jbdc0vTOcol

[17] Paul C. Tetlock. 2007. Giving Content to Investor Sentiment: The Role of Media in the Stock Market. *The Journal of Finance* 62, 3 (2007), 1139–1168. https://doi.org/10.1111/j.1540-6261.2007.01232.x

A ARTIFACT ROOTS AND REPLICATION COMMANDS

Repository: https://github.com/ValdiviaEng/financial-event-state-forecasting.git. Primary evidence roots are XGBoost data/v1p0_full_results/v1p0__phase1_top50_updated_av__stage43__xgboost_tuning_20260528T100000Z__src_6dff1bd__run_40f1c9a0/, the Mixed model data/v1p0_full_results/v1p0__phase1_top50_updated_av__stage48__frozen_tsmixer_horizon_mixer__20260529T093301Z__src_6dff1bd__run_fc4c1756/, and the lockbox portfolio simulation data/v1p0_full_results/v1p0__phase1_top50_updated_av__stage47__lockbox_portfolio_simulation__20260529T095157Z__src_6dff1bd__run_8317929b/. Validation command: `source ~/venvs/ai/bin/activate && python -m pytest tests`.

B EXACT LOCKBOX RESULTS

Table 2: Lockbox metrics by model and horizon. Lower RMSE/MAE is better; R^2 is against the zero-return baseline.

Model	Horizon	RMSE	MAE	R^2
XGBoost	1d	0.0214	0.0146	0.0000
TSMixer	1d	0.0221	0.0154	-0.0640
Mixed model	1d	0.0214	0.0146	-0.0000
XGBoost	5d	0.0468	0.0335	0.0000
TSMixer	5d	0.0469	0.0337	-0.0085
Mixed model	5d	0.0469	0.0336	-0.0058
XGBoost	10d	0.0661	0.0488	0.0000
TSMixer	10d	0.0670	0.0494	-0.0262
Mixed model	10d	0.0665	0.0490	-0.0101
XGBoost	20d	0.0943	0.0696	-0.0014
TSMixer	20d	0.0949	0.0701	-0.0148
Mixed model	20d	0.0944	0.0696	-0.0031
XGBoost	60d	0.1929	0.1396	0.0072
TSMixer	60d	0.1972	0.1398	-0.0374
Mixed model	60d	0.1942	0.1384	-0.0063

Table 3: Best portfolio row by horizon over the 137-day lockbox interval. The rows summarize realized forecast-ranking simulations under the stated frictionless assumptions.

Horizon	Model / policy	Return	Final equity	Trades
1d	TSMixer / one_d_bot10	15.05%	$115,045	1161
5d	TSMixer / one_d_bot10	5.24%	$105,240	1831
10d	TSMixer / none	9.48%	$109,482	1780
20d	TSMixer / none	7.54%	$107,536	1680
60d	TSMixer / botk_3_of_5	12.43%	$112,434	1319

C SUPPLEMENTARY VISUALIZATIONS

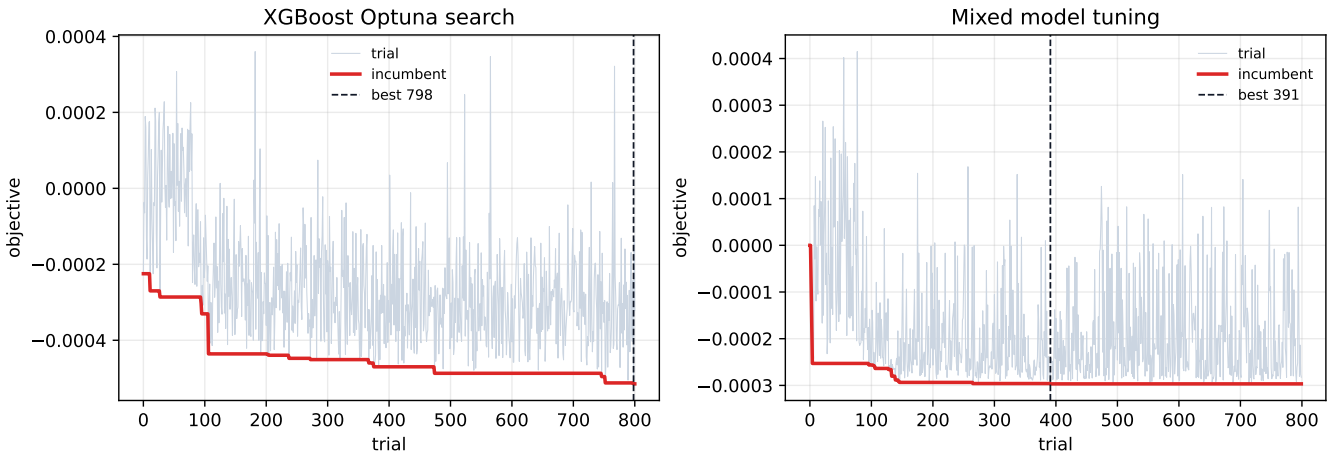


Figure 2: Optuna incumbent curves for XGBoost and the Mixed model.

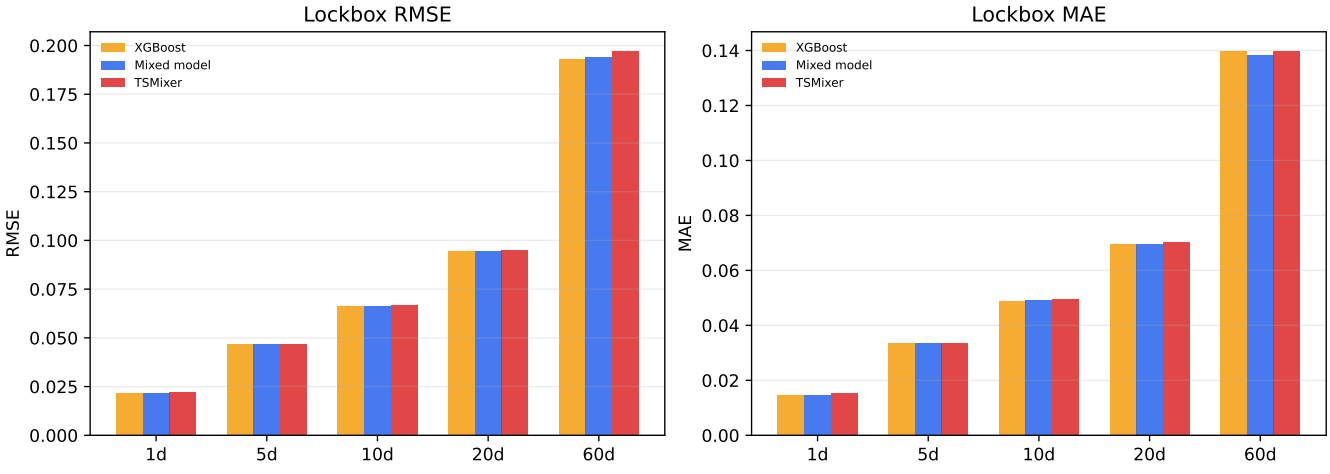


Figure 3: Lockbox RMSE and MAE by horizon for XGBoost, TSMixer, and the Mixed model.

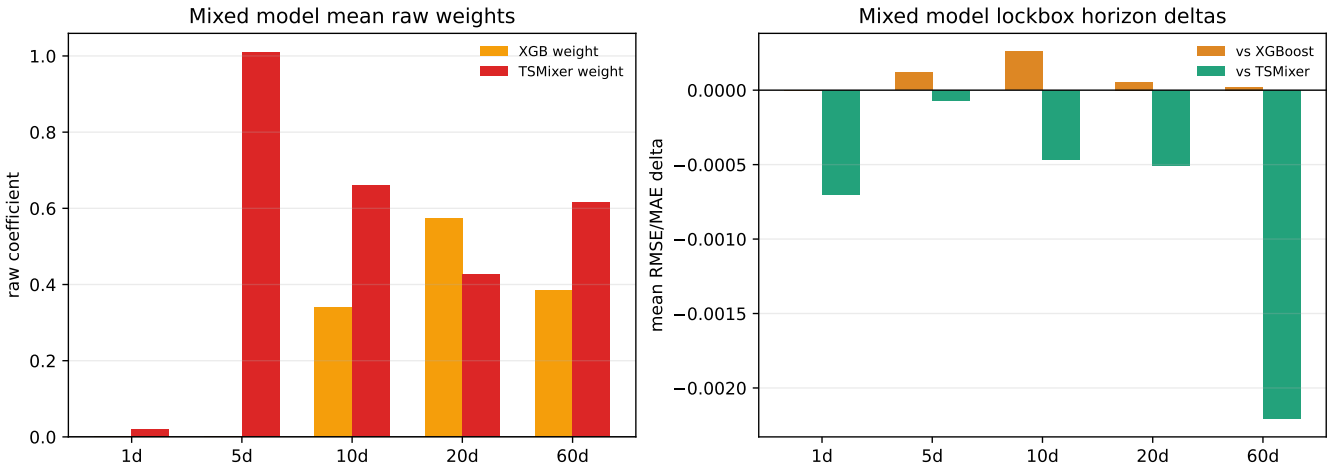


Figure 4: Horizon-specific raw mixer weights and lockbox horizon deltas.

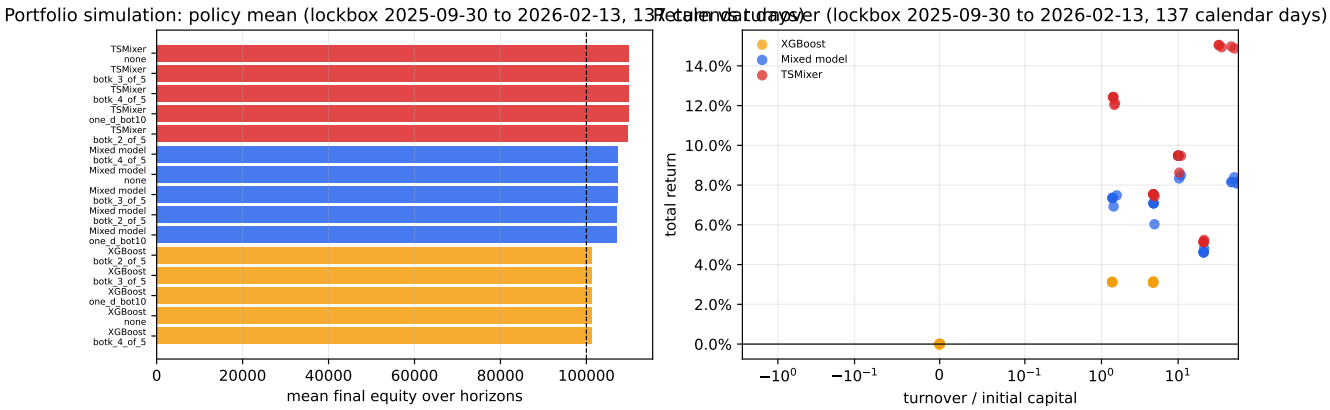


Figure 5: Portfolio simulations over the 2025-09-30 to 2026-02-13 lockbox interval (137 calendar days): final equity by policy and return versus turnover.

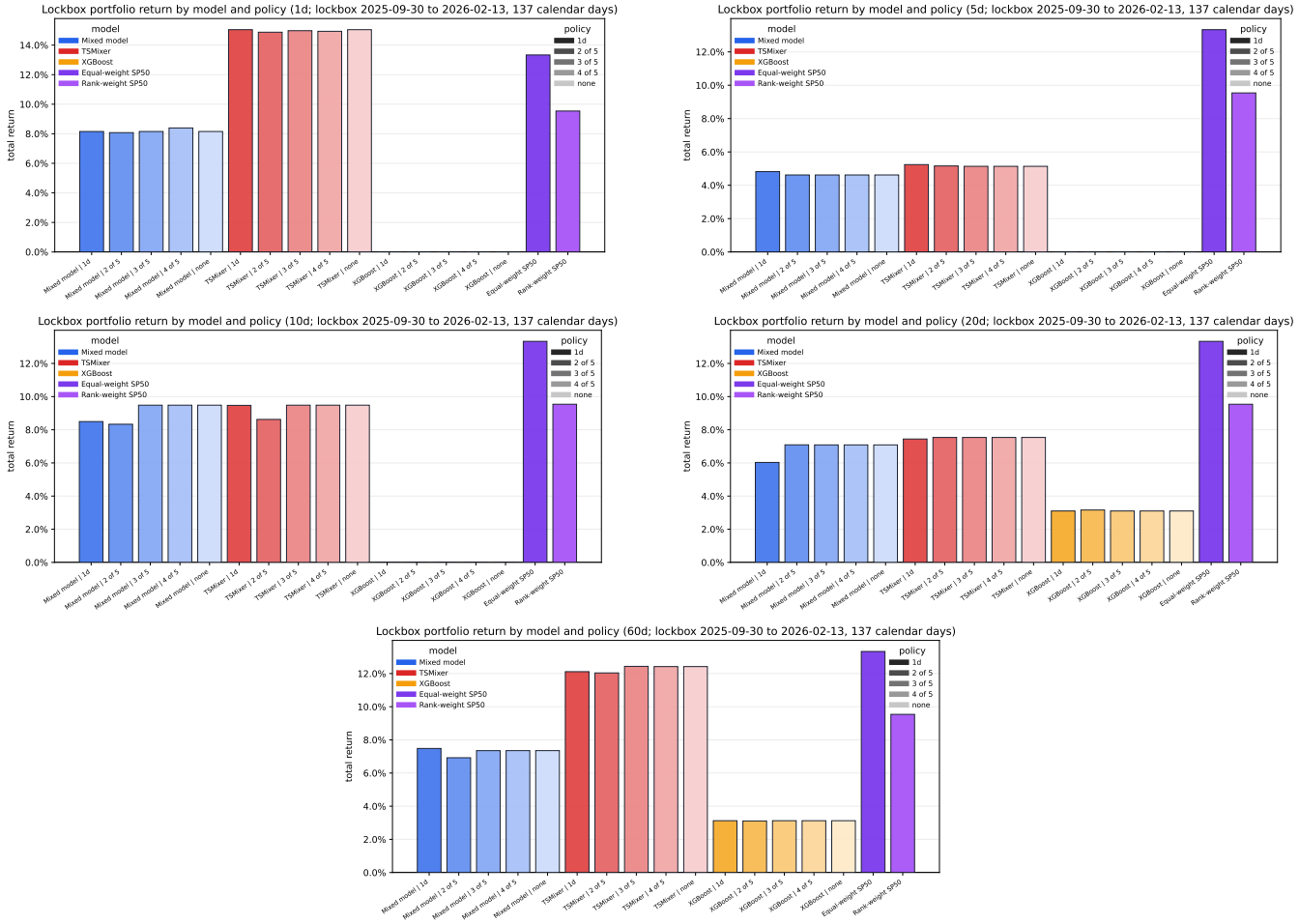


Figure 6: Portfolio return bars by horizon, model, sell policy, and SP50 benchmark proxy over the 137-day lockbox interval.

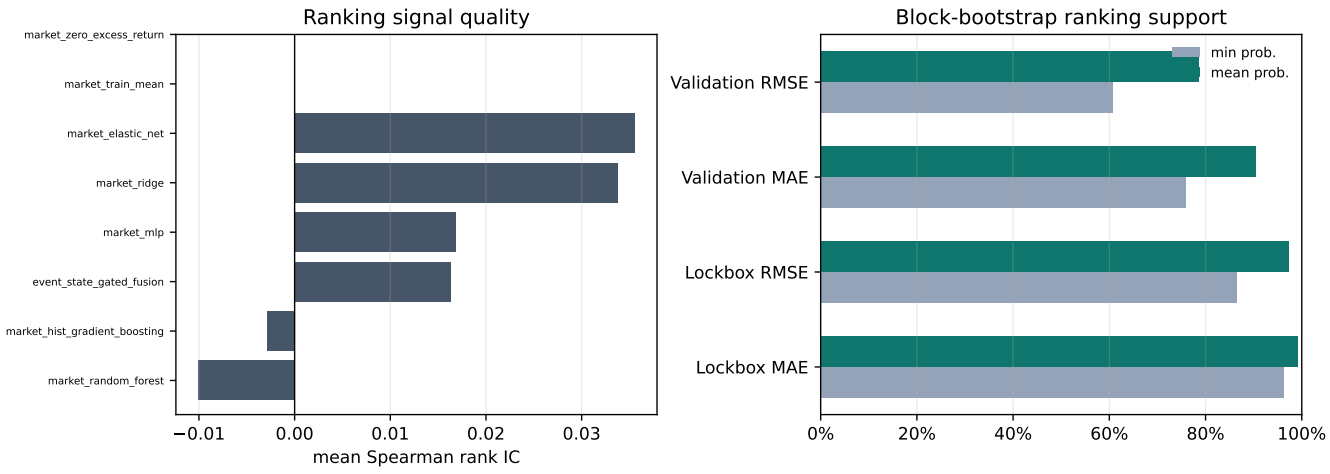


Figure 7: Ranking signal quality and block-bootstrap support.